Answer Key to Problem Set 1

Econ 5030 — 2026

1 Preferences and Utility

1. Assume that $\succsim$ is complete and transitive on X. Show the following properties must hold for $\sim$ and $\succ$.

   (a) $x \succ x$ never holds.

   Proof: By definition, $x \succ y$ means that $x \succsim y$ and not $y \succsim x$. Therefore, $x \succ x$ would require both $x \succsim x$ and not $x \succsim x$, which is impossible.

   (b) If $x \succ y$ and $y \succ z$, then $x \succ z$.

   Proof: Since $x \succ y$ and $y \succ z$, we have $x \succsim y$ and $y \succsim z$. Transitivity implies $x \succsim z$. It remains to show that not $z \succsim x$. Suppose instead that $z \succsim x$. Then $z \succsim x$ and $x \succsim y$ imply $z \succsim y$, contradicting $y \succ z$. Hence, not $z \succsim x$, and therefore $x \succ z$.

   (c) $x \sim x$ for all x.

   Proof: Completeness applied to the pair (x, x) implies $x \succsim x$. This single statement supplies both conditions in the definition of indifference when the two alternatives are the same. Hence, $x \sim x$.

   (d) If $x \sim y$ and $y \sim z$, then $x \sim z$.

   Proof: From $x \sim y$, we have $x \succsim y$ and $y \succsim x$. From $y \sim z$, we have $y \succsim z$ and $z \succsim y$. Transitivity gives $x \succsim z$ and $z \succsim x$. Hence, $x \sim z$.

   (e) If $x \sim y$, then $y \sim x$.

   Proof: By definition, $x \sim y$ means $x \succsim y$ and $y \succsim x$. These are exactly the two conditions required for $y \sim x$.

   (f) If $x \succ y$ and $y \succsim z$, then $x \succ z$.

   Proof: Since $x \succ y$, we have $x \succsim y$ and not $y \succsim x$. Together with $y \succsim z$, transitivity gives $x \succsim z$. Suppose that $z \succsim x$. Then $y \succsim z$ and $z \succsim x$ imply $y \succsim x$, a contradiction. Therefore, not $z \succsim x$, and hence $x \succ z$.

2. Take the set $X = \{a, b, c, d\}$. Suppose that $\succsim$ is complete and transitive on X.

(a) Prove that the set of optimal choices

$$C(X, \succsim) = \{x \in X : x \succsim y \text{ for all } y \in X\}$$

is non-empty.

Proof: We construct an optimal element by comparing alternatives one at a time. Begin with a and compare it with b. By completeness, either $a \succsim b$ or $b \succsim a$. Let w_2 be an alternative that is weakly preferred to both a and b.

Next compare w_2 with c. By completeness, one of these two alternatives is weakly preferred to the other. Call it w_3. By construction, $w_3 \succsim w_2$ and $w_3 \succsim c$, so transitivity implies that w_3 is weakly preferred to a, b, and c.

Finally, compare w_3 with d and call the weakly preferred alternative w_4. By the same argument, $w_4 \succsim y$ for every $y \in X$. Therefore, $w_4 \in C(X, \succsim)$, so $C(X, \succsim)$ is non-empty.

(b) Prove that $x \sim y$ for all $x, y \in C(X, \succsim)$.

Proof: Let $x, y \in C(X, \succsim)$. Since x is optimal, $x \succsim y$. Since y is optimal, $y \succsim x$. Therefore, $x \sim y$ by the definition of indifference.

3. Take the set $X = \{a, b, c, d\}$.

(a) Prove that if $\succsim$ is complete and transitive on X, then there exists a function $u : X \to \mathbb{R}$ that represents the preferences $\succsim$ on X.

Proof: Indifference is an equivalence relation: it is reflexive, symmetric, and transitive by Problem 1. Therefore, it partitions X into indifference classes. Completeness and transitivity order these classes from worst to best.

Because X is finite, there are finitely many indifference classes. Denote them, from worst to best, by $E_1, \ldots, E_K$, where $1 \leq K \leq 4$. Define

$$u(x) = k \qquad \text{if } x \in E_k.$$

If $x \succsim y$, then the class containing x is ranked at least as high as the class containing y, so $u(x) \geq u(y)$. Conversely, if $u(x) \geq u(y)$, then the class containing x is ranked at least as high as the class containing y, so $x \succsim y$. Hence,

$$x \succsim y \quad \iff \quad u(x) \geq u(y),$$

and u represents $\succsim$.

(b) Explain why we need an additional assumption to obtain a utility representation if $X' = [0,1]^2$. Give an example of preferences that are complete and transitive on X' but cannot be represented by a real-valued utility function.

Answer: On an infinite choice set, completeness and transitivity alone do not guarantee a real-valued utility representation. A standard additional assumption is continuity.

Consider lexicographic preferences on $[0,1]^2$. Define

$$(x_1, x_2) \succ_L (y_1, y_2)$$

if either $x_1 > y_1$, or $x_1 = y_1$ and $x_2 > y_2$. These preferences are complete and transitive.

Suppose that a real-valued function u represented them. For every $a \in [0,1]$, we have $(a,1) \succ_L (a,0)$, so

$$u(a,1) > u(a,0).$$

Thus, the open interval

$$I_a = \big(u(a,0), u(a,1)\big)$$

is non-empty. If $a < b$, then $(b,0) \succ_L (a,1)$, so $u(b,0) > u(a,1)$. Therefore, the intervals I_a and I_b are disjoint.

Every non-empty open interval contains a rational number. Choosing one rational number from each I_a would assign distinct rational numbers to the uncountably many values of $a \in [0,1]$. This is impossible because the rational numbers are countable. Hence, lexicographic preferences cannot be represented by a real-valued utility function.

4. Consider the utility-maximization problem on $X = \mathbb{R}^2_+$ with Cobb–Douglas utility

$$u(x_1, x_2) = x_1^{1/2} x_2^{1/2},$$

where $p = (p_1, p_2) \in \mathbb{R}^2_{++}$ and $w > 0$.

(a) Show that an optimal bundle is interior and exhausts the budget. Use the first-order conditions to derive the Marshallian demand $D(p,w)$ and the indirect utility function $v(p,w)$.

Answer: First, an optimum must be interior. Any bundle with $x_1 = 0$ or $x_2 = 0$ gives utility zero, while a feasible bundle with both goods strictly positive gives strictly positive utility. In addition, utility is strictly increasing in both goods when $x_1, x_2 > 0$. Therefore, leaving

income unspent cannot be optimal, and the budget constraint binds:

$$p_1 x_1 + p_2 x_2 = w.$$

At an interior solution, maximizing u is equivalent to maximizing its logarithm. The Lagrangian is

$$\mathcal{L} = \frac{1}{2}\log x_1 + \frac{1}{2}\log x_2 + \lambda(w - p_1 x_1 - p_2 x_2).$$

The first-order conditions are

$$\begin{aligned} \frac{1}{2x_1} &= \lambda p_1, \\ \frac{1}{2x_2} &= \lambda p_2. \end{aligned}$$

Dividing the two conditions gives

$$\frac{x_2}{x_1} = \frac{p_1}{p_2},$$

or, equivalently, $p_1 x_1 = p_2 x_2$. Thus, the consumer spends one-half of income on each good. Combining this condition with the binding budget constraint gives the Marshallian demand

$$D(p, w) = \left\{ \left(\frac{w}{2p_1}, \frac{w}{2p_2} \right) \right\}.$$

Substituting the demanded bundle into the utility function gives the indirect utility function

$$\begin{aligned} v(p, w) &= \left(\frac{w}{2p_1} \right)^{1/2} \left(\frac{w}{2p_2} \right)^{1/2} \\ &= \frac{w}{2\sqrt{p_1 p_2}}. \end{aligned}$$

(b) Compute the demanded bundle and indirect utility when $(p_1, p_2, w) = (4, 1, 80)$.

Answer: Substituting $(p_1, p_2, w) = (4, 1, 80)$ gives

$$x_1^* = \frac{80}{2(4)} = 10, \qquad x_2^* = \frac{80}{2(1)} = 40.$$

Therefore,

$$D(4, 1, 80) = \{(10, 40)\} \qquad \text{and} \qquad v(4, 1, 80) = \sqrt{10 \cdot 40} = 20.$$

(c) Verify Walras' law and degree-zero homogeneity of demand at the values in part (b), using $\lambda = 2$.

Answer: Walras' law holds because expenditure equals income:

$$4(10) + 1(40) = 80 = w.$$

To check degree-zero homogeneity, double all prices and income. Then $(2p_1, 2p_2, 2w) = (8, 2, 160)$, and

$$D(8, 2, 160) = \left\{ \left(\frac{160}{2(8)}, \frac{160}{2(2)} \right) \right\} = \{(10, 40)\} = D(4, 1, 80).$$

Thus, doubling prices and income leaves demand unchanged, as degree-zero homogeneity requires.

5. Consider the utility-maximization problem on $X = \mathbb{R}^2_+$ with perfect-complements utility

$$u(x_1, x_2) = \min\{x_1, 2x_2\},$$

where $p = (p_1, p_2) \in \mathbb{R}^2_{++}$ and $w \geq 0$.

(a) Derive the Marshallian demand $D(p, w)$ and the indirect utility function $v(p, w)$. Give an economic explanation for the proportions in the optimal bundle.

Answer: Let t denote the utility level the consumer wants to attain. Attaining utility of at least t requires

$$x_1 \geq t \qquad \text{and} \qquad x_2 \geq \frac{t}{2}.$$

Because both prices are strictly positive, the least expensive bundle that attains utility t is

$$(x_1, x_2) = \left(t, \frac{t}{2} \right).$$

Its cost is

$$p_1 t + p_2 \frac{t}{2} = \left(p_1 + \frac{p_2}{2} \right) t.$$

The largest affordable utility level therefore satisfies

$$\left(p_1 + \frac{p_2}{2} \right) t^* = w,$$

so

$$t^* = \frac{w}{p_1 + p_2/2} = \frac{2w}{2p_1 + p_2}.$$

It follows that

$$D(p,w) = \left\{ \left(\frac{2w}{2p_1 + p_2}, \frac{w}{2p_1 + p_2} \right) \right\}$$

and

$$v(p,w) = \frac{2w}{2p_1 + p_2}.$$

Economically, the two goods are useful only in the fixed proportion $x_1 = 2x_2$. Extra units of either good by itself do not increase utility. Since prices are positive, purchasing such excess units would waste income, so the consumer chooses the kink of the indifference curve.

(b) Compute the demanded bundle and indirect utility when $(p_1, p_2, w) = (3, 2, 80)$.

Answer: Here $2p_1 + p_2 = 2(3) + 2 = 8$. Therefore,

$$x_1^* = \frac{2(80)}{8} = 20, \qquad x_2^* = \frac{80}{8} = 10.$$

Thus,

$$D(3,2,80) = \{(20,10)\} \qquad \text{and} \qquad v(3,2,80) = \min\{20, 2(10)\} = 20.$$

The bundle exhausts the budget:

$$3(20) + 2(10) = 80.$$

(c) Show that the preferences represented by u are convex but not strictly convex. Explain why the demand correspondence is nevertheless a singleton (and hence convex-valued).

Answer: For any utility level $t \geq 0$, the upper contour set is

$$\begin{aligned} U(t) &= \{(x_1, x_2) : \min\{x_1, 2x_2\} \geq t\} \\ &= \{(x_1, x_2) : x_1 \geq t \text{ and } x_2 \geq t/2\}. \end{aligned}$$

This set is the intersection of two half-spaces and is therefore convex. Hence, the preferences are convex.

The preferences are not strictly convex. For $x = (1,1)$ and $y = (1,2)$, we have $x \neq y$ and $u(x) = u(y) = 1$, but $u\left(\frac{1}{2}x + \frac{1}{2}y\right) = u(1, 3/2) = 1$. Thus, the midpoint is indifferent to x and y, rather than strictly preferred to them.

Nevertheless, demand is a singleton. At utility t, strictly positive prices make $(t, t/2)$ the unique least-cost bundle, and the budget fixes a unique maximal t. Hence, the demand set is a convex singleton.